ESP-IDF Trace Configuration Code — Full Trace Explanation

This document contains the exact, full explanation of the ESP-IDF trace configuration function, describing its runtime behavior, memory lifetime, and trace data flow.

```
esp_trace_open_params_t esp_trace_get_user_params(void)
{
    static esp_apptrace_config_t app_trace_config = APPTRACE_CONFIG_DEFAULT();
#if CONFIG_APPTRACE_DEST_UART
    /* Override default values to use console pins as a uart channel */
    app_trace_config.dest_cfg.uart.tx_pin_num = U0TXD_GPIO_NUM;
    app_trace_config.dest_cfg.uart.rx_pin_num = U0RXD_GPIO_NUM;
#endif

    esp_trace_open_params_t trace_params = {
        .core_cfg = NULL,
        .encoder_name = "sysview",
        .encoder_cfg = NULL,
        .transport_name = "apptrace",
        .transport_cfg = &app_trace_config,
    };
    return trace_params;
}
```

Function signature

The function returns an esp_trace_open_params_t structure. This structure tells ESP-IDF how trace data should be encoded and transported off the device.

Static AppTrace configuration

A static esp_apptrace_config_t variable is initialized using APPTRACE_CONFIG_DEFAULT(). The static keyword ensures the structure remains valid after the function returns, which is required because a pointer to it is stored inside the returned trace parameters.

Conditional UART override

When CONFIG_APPTRACE_DEST_UART is enabled, the code overrides the default UART pin assignments and explicitly selects UART0 TX and RX pins. This routes trace output over the ESP console UART.

Trace open parameters

The esp_trace_open_params_t structure defines the complete trace pipeline. core_cfg is set to NULL, meaning default tracing behavior is used for all CPU cores.

encoder_name is set to "sysview", selecting the SEGGER SystemView encoder. encoder_cfg is NULL, so default encoder settings are used.

transport_name is set to "apptrace", selecting ESP AppTrace as the transport. transport_cfg points to the static AppTrace configuration structure.

Runtime trace data flow

At runtime, ESP-IDF generates trace events which are encoded using the SystemView format and transmitted off the chip using AppTrace over the configured interface. A host tool such as SEGGER SystemView receives and visualizes the data.

Summary

This function constructs and returns a complete and safe trace configuration. It combines SystemView encoding with AppTrace transport and ensures correct memory lifetime through static configuration storage.